

Mars Orbiter Mission (Mangalyaan)

Indian Space Research Organisation (ISRO) — 2013–2022

1. Mission Overview

The Mars Orbiter Mission (MOM), informally known as Mangalyaan, is India's first interplanetary mission. It was launched on 5 November 2013 by the Polar Satellite Launch Vehicle (PSLV-C25) from Satish Dhawan Space Centre, Sriharikota. The spacecraft was inserted into Mars orbit on 24 September 2014, making India the first Asian nation to reach Martian orbit and the first nation in the world to do so in its maiden attempt. The total mission cost was approximately INR 450 crore (approximately USD 74 million at the time), making it the least expensive Mars mission to date.

2. Mission Objectives

The primary objective of Mangalyaan was to develop the technologies required for design, planning, management, and operations of an interplanetary mission. The secondary scientific objectives included exploration of Mars surface features, morphology, mineralogy, and the Martian atmosphere using indigenous scientific instruments. The mission demonstrated ISRO's capability to develop and operate a spacecraft over interplanetary distances.

3. Scientific Instruments

Mangalyaan carries five scientific instruments with a total payload mass of 15 kg. The instruments are: LAP (Lyman Alpha Photometer) for measuring the relative abundance of deuterium and hydrogen in the Martian upper atmosphere; MSM (Methane Sensor for Mars) for measuring methane in the Martian atmosphere; TIS (Thermal Infrared Imaging Spectrometer) for mapping surface composition and mineralogy; MCC (Mars Colour Camera) for taking images of the Martian surface and its moons Phobos and Deimos; and MENCA (Mars Exospheric Neutral Composition Analyser) for studying the neutral composition of the Martian exosphere.

4. Spacecraft and Orbit

The Mangalyaan spacecraft has a launch mass of 1337 kg including 852 kg of propellant. It uses a 440 N liquid apogee motor for major manoeuvres. The spacecraft is in a highly elliptical orbit around Mars with a periapsis of approximately 421 km and an apoapsis of approximately 76,993 km, with an orbital period of about 72.7 hours. The orbit was chosen to allow observation of the entire Martian disc and to study the upper atmosphere.

5. Journey to Mars

After launch on 5 November 2013, Mangalyaan performed a series of orbit-raising manoeuvres around Earth before the Trans-Mars Injection (TMI) burn on 1 December 2013. The spacecraft travelled approximately 780 million km over 298 days to reach Mars. The Mars Orbit Insertion (MOI) burn was performed on 24 September 2014, lasting approximately 24 minutes, reducing the spacecraft velocity sufficiently to be captured by Martian gravity. The mission operated for over 8 years before communication was lost in September 2022.